

SWARM-PROJECT II

Decentralized cognitive swarm for search-and-rescue & intercept — five multirotors, no central brain, real flight physics.

Essence

Each drone is an intelligent actor: it bids on tasks using distance + wind-aware energy belief, vetoes unsafe work, relinquishes and hands over when depleted, and pursues moving targets from shared data. Coordination is a replicated, elected auction that survives node death. Everything runs on ArduPilot physics today and is shaped for real hardware tomorrow.

Use cases

Case	Trigger	Swarm behavior
Flood / disaster SAR	YOLO person sighting → human confirm	nearest eligible drone dispatched; handover on low battery
Perimeter / static intercept	one click	simultaneous lift, rotating 300 m ring, FORMED banner, auto-release
Moving-target pursuit	one click (mover auto-starts)	ring chases live track; lead cutters; stale-track hold

Technical highlights

- ROS 2 Jazzy over **Zenoh** (mesh-ready, peer mode)
 - Contract-net auction, **replicated + elected leader**, kill-tolerant
 - Namespaced perception /swarm/Dx/camera; **detections-only** shared bus
- Mission-state events: PERIMETER_FORMED / DEGRADED / RELEASED
 - 2D+3D GCS: X500 airframes, 5 Hz attitude-first, quarry + live ring
 - Regression gate: 6 acceptance criteria + graphs + run history

Status & roadmap

DONE

fleet+mesh+auction · SAR/HITL · wind budgets · self-healing coordination · perimeter · pursuit · namespaced perception · regression gate

NEXT

hardware P1-P4 (1 real + 4 SITL → full fleet) · safety pack · radio realism (NS-3) · CesiumJS photoreal · advisory NNs

Hardware plan

Pixhawk + Jetson Orin Nano + CSI cam + mesh radio per drone; video pulled selectively (RTSP/WebRTC), telemetry on Zenoh; phased bench → hybrid → full-fleet field mesh.